MARKING SCHEME

NETWORK DESIGN AND MANAGEMENT

CAT 3,TERM 3

Questions 1 to 5 (2 marks each)

Q	Expected answer	Marks
1	The process of checking a network's hardware, cabling, connections and software (1) to confirm it is installed correctly, performs as designed and is free of faults (1).	2
2	Any two, 1 mark each: detects faults early; confirms correct installation to standards; ensures expected performance; speeds up troubleshooting; saves cost and time; improves security; supports documentation and warranty claims; helps capacity planning; ensures electrical safety.	2
3	Clamp meter (1). Connected in parallel, across the two points being tested (1).	2
4	Npcap is the packet capture library or driver that sits between Wireshark and the network adapter (1). Without it Wireshark opens but cannot capture packets from the interface (1).	2
5	A logical grouping of devices regardless of their physical location (1). It divides a switch into separate broadcast domains (1).	2

Question 6 (6 marks)

Part	Expected answer	Marks
(a)	TTL is a field in the IP header that limits the number of hops a packet may make (1). At 0 the router drops the packet and sends back an ICMP Time Exceeded message (1).	2
(b)	128 - 9 = 119 (1). Not dropped, because the TTL is still above 0 (1).	2
(c)	Tracert sends probes with TTL 1, then 2, then 3, increasing by 1 each time (1). Each router that reduces the TTL to 0 drops the probe and returns ICMP Time Exceeded, revealing its address and hop position (1).	2

Question 7 (6 marks)

Part	Expected answer	Marks
(a)	Hop 4 (1). A slow or congested link between hop 3 and hop 4 (1).	2
(b)	Any two, 1 mark each: the router is configured to block or ignore ICMP; a firewall drops the probes; the router gives low priority to replying to probes; temporary packet loss. Since hop 6 replied, the path itself is working.	2
(c)	Ping only shows whether the destination is reachable and the total round-trip time (1). It does not list each router, so it cannot show at which hop the delay begins (1).	2

Question 8 (6 marks)

Part	Expected answer	Marks
(a)	Step 1: the client sends SYN (1). Step 2: the server replies with SYN-ACK (1). Step 3: the client sends ACK to confirm, and the connection is established (1). Flags: SYN=1, ACK=0 in step 1; SYN=1, ACK=1 in step 2; SYN=0, ACK=1 in step 3 (1, all three correct).	4
(b)(i)	tcp.flags.syn == 1 && tcp.flags.ack == 0	1
(b)(ii)	icmp.type == 8	1

Question 9 (6 marks)

Finding	Interpretation (1 mark)	Action (1 mark, any suitable)
(i)	180 V is far below the expected 240 V, indicating a supply problem that could damage equipment.	Report to an electrician or the power supplier, use a voltage stabiliser or UPS, and do not connect sensitive equipment until corrected.
(ii)	Pin 5 is open, caused by a broken wire or a poor crimp.	Re-terminate the connector (or replace the cable) using T568B and retest.
(iii)	-82 dBm is a very weak signal, so the connection will be slow or unreliable.	Move the access point closer, add another access point, or change the channel to reduce interference, then re-measure.

Question 10 (6 marks)

Part	Expected answer	Marks
(a)	Create the VLAN: vlan 10 and name FINANCE (1). Select the port: interface fa0/1 and switchport mode access (1). Assign the VLAN: switchport access vlan 10 (1). Accept enable and configure terminal as part of the answer.	3
(b)	The two PCs are in different VLANs, which are separate broadcast domains, so the switch does not forward traffic between them (1). A Layer 3 device, such as a router or Layer 3 switch, is needed for inter-VLAN routing (1). Example: router-on-a-stick, with the switch port to the router set as a trunk, router subinterfaces using encapsulation dot1Q with IP addresses 192.168.10.1 and 192.168.20.1, and these set as the default gateways of the PCs (1).	3

Mark Distribution Summary

Question	Marks
1 to 5 (2 marks each)	10
6	6
7	6
8	6
9	6

Question	Marks
10	6
Total	40

Award marks for any correct, relevant alternative answer.